Document classification and claim extraction

AI service: documents pipeline · Run 09 Nov 2026, 00:55 UTC · **All targets met**

What is measured

Whether each claim document is recognized as the right type, and whether every field and bill line is read correctly with a quote on the page. A field the document does not contain must come back as missing, never invented. Text inside a document that tries to instruct the model must have no effect.

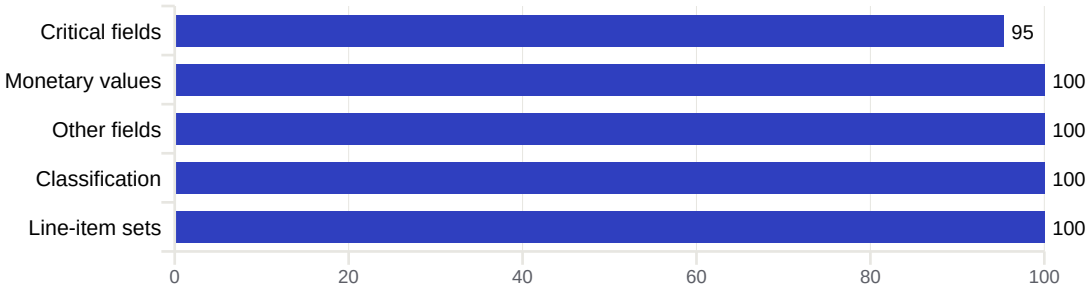
Data

17 synthetic claim documents with known answers: clean PDFs, scans, a two-page TIFF, handwriting-style pages, dense and sparse bills, conflicting documents, and documents carrying injected instructions.

Results

Measure	Result	Target	
Critical field accuracy	**61 / 64 (95.3%)**	≥ 95%	**Met**
Monetary value accuracy	**8 / 8 (100.0%)**	≥ 98%	**Met**
Values invented for absent fields	**0**	0	**Met**
Injected values returned	**0**	0	**Met**
Other field accuracy	**28 / 28 (100.0%)**		
Document type classification	**17 / 17 (100.0%)**		
Bill line items read completely	**8 / 8 (100.0%)**		
Documents that failed	**0**	0	**Met**
Model calls per document	**2.4**		
Cost per document	**USD 0.0012**		

Accuracy by measure (%)



Observations

• Model deepseek/deepseek-v4.1-flash-1017, escalating to anthropic/claude-haiku-5.5 when a check fails. Pipeline doc-1.
• Error: WRONG claims/scanned/discharge-note.tiff:admission_date: expected '2026-03-14', got UNCERTAIN '2026-03-14' raw='14/03/2026' reason=REREAD_DISAGREES
• Error: WRONG claims/scanned/discharge-note.tiff:discharge_date: expected '2026-03-17', got UNCERTAIN '2026-03-17' raw='17/03/2026' reason=REREAD_DISAGREES
• Error: WRONG claims/scanned/discharge-note.tiff:patient_name: expected 'Synthetic Patient Asha', got UNCERTAIN 'Synthetic Patient Asha' raw='Synthetic Patient Asha' reason=REREAD_DISAGREES

• A value read wrongly but marked uncertain is not shown as fact: it reaches the reviewer as an open question.

Limitations

• Synthetic documents only. Real hospital paperwork is more varied.
• One run. Model output varies slightly between runs.

Reproduce

```
cd services/ai; python -m uv run python evals/run_claim_extraction.py
```